Arpita Bhattacharya

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I am a curious and enthusiastic Masters in Mathematics and Numerical Analysis from Lund University, Sweden, highly motivated and excited to apply my academic and computational experience to solve challenging problems in multiple fields of science and technology. I am especially interested to use my analytical mindset to further study and explore the field of applied mathematics, especially Dynamical Systems, Analysis and Computational Mathematics. I am eager to contribute my skills and knowledge for further growth and meaningful research work in the field while continuing to develop as a professional.

Research and Project Experience

Master Thesis (Supervisor: Prof. Erik Wahlén) — February 2025 – March 2026

Title: *Singularities in Steady Water Waves using Analytic Continuation*
Study of the analytic structure of steady two-dimensional water waves using tools from complex analysis, conformal mapping, and analytic continuation. The focus was on the study of the singularities in analytically continued wave profiles and how these influence the regularity and convergence properties of free-surface water wave representations.

Projects and Relevant Courses (Master and Bachelor) — September 2023 – January 2025

Analysis and Differential equations

Partial and Ordinary Differential Equations
Analyze and solve dynamical systems using differential equations including methods of mathematical physics. Connected the theoretical systems to physical phenomena such as heat, waves, diffusion, transport and stability.

Analytic Functions, Complex and Real Analysis
In depth study of complex and real functions, convergence, singularities, transformations and related theorems.

Integration Theory, Multivariable Calculus and Optimization
Strong foundation in measure theory and problems based in higher dimensions. Clear understanding of mathematical optimization problems.

Computational Mathematics and Numerical Analysis

Mathematical Modelling (Under guidance of: *Prof. Joachim Hein*, Lund University, Sweden)
Linear optimization problems implemented using Python and Microsoft Excel.
Project 1: *Distribution Optimization*: Minimize cost of Textile distribution
Project 2: *Predator-prey modelling*: Analyze population coexistence of species

Numerical Methods for Differential Equations (Under guidance of: *Prof. Tony Stillfjord*, Lund University, Sweden)
Implementation of projects in MATLAB. The projects used concepts such as: Euler method, Adaptive Runge-Kutta methods, Lotka-Volterra equation, Van der Pol problem, 2-point BVP solver, Beam equation, Strum-Liouville eigenvalue problem, Schrödinger equation, Diffusion equation, Advection equation, Convection diffusion equation, Viscous Burgers' equation.

Numerical Methods for Partial Differential Equations (Under guidance of: *Prof. Robert Klöfkorn*, Lund University, Sweden)
Projects based on Elliptic partial differential equations, investigation of their properties using DUNE and DUNE-FEM in Python.

Numerical Linear Algebra
Implementation of project tasks/assignments in Python and MATLAB, using concepts such as: Gram-Schmidt orthogonalization, QR-factorization, Householder transformations, Givens rotations, Image Compression using SVD, Hilbert Matrices, LUdecomposition, hyperlink matrices, GMRES.

Other computational bachelor experiences
Algorithm structure, graphing and numerical solutions using MATLAB. Scientific software development with computation results using Java.

Algebra and Number Theory

Number Theory - ElGamal Cryptography
Implementation of the ElGamal cryptosystem in Python. Submitted report demonstrating the working of the algorithm.

Elliptic Curve Cryptography (Under guidance of: *Dr. Weiyi Zhang*, University of Warwick, United Kingdom)
A research essay on the cryptography and applications using Abstract Algebra and Number Theory.

Other relevant courses: Galois Theory, Group and representation theory, Algebraic Number Theory, Abstract and Advanced Linear Algebra

EDUCATION

Lund University Lund, Sweden	**MASTER IN MATHEMATICS** (Numerical Analysis)	Sep 2023 – Mar 2026	Grade: G
University of Warwick Coventry, United Kingdom	**BACHELOR IN MATHEMATICS**	Sep 2019 – Aug 2022	Grade: Hons.
National Public School Bengaluru, Karnataka, India	**HIGH SCHOOL**	Jun 2005 – Mar 2019	Final Score: 91%
Centre of Fundamental Research and Creative Education (CFRCE) Bengaluru, Karnataka, India	**RESEARCH EDUCATION**	Apr 2017 – Aug 2019	GPA: 3.7

SKILLS

Software Programming	Python, Java, R, C++	★★★☆☆
Simulation	MATLAB	★★★☆☆
ML/AI	PyTorch	★☆☆☆☆
Data Processing/Visualization	Tableau	★★☆☆☆
Documentation	Markdown, Latex, MS Office	★★★★☆

EXTRA-CURRICULAR

International Student Ambassador Lund University, Lund, Sweden — September 2023 – March 2026

- Responsible for tasks designed to help prospective students in their decision-making process.
- Be a representative of the University.
- Answer questions and provide information via the Unibuddy platform and Live events.
- Create content for blogs and social channels of the University.

Company Host at ARKAD (Career Fair) Lund University, Lund, Sweden — October 2023

- Point of contact between ARKAD and visiting companies in the career fair. Help order resources and materials, set up booths, personally assist the company representatives.

Warwick Student Volunteer University of Warwick, Coventry, United Kingdom — January 2020 – March 2020

- Helped primary school children learn mathematics through fun games, building their confidence in math concepts and making it fun as part of project 'Fun with numbers'.

National Centre for Human Settlements and Environment (NCHSE) MP, India — April 2018

- As part of "Integrated Watershed Management Programme" of the NCHSE, survey health, hygiene and sanitation awareness in the villages within the districts of Madhya Pradesh, India

OTHER INTERESTS/HOBBIES

Music

INSTRUMENT	▪ Sitar – Hindustani Classical
DANCE	▪ Hindustani Classical – Kathak, Bharat Natyam ▪ Other styles – Contemporary, Street, Hip-hop

Reading: FICTION, SCIENCE, TECHNOLOGY, PHILOSOPHY